Technical Report: Healthcare Data Ingestion and Analysis using Amazon S3 and Amazon Athena

1. Objective

To document the process of building a cloud-based storage and query environment for healthcare data, from raw data ingestion to data extraction and analysis using SQL queries via Amazon Web Services (AWS).

2. Technologies & Services Used

- **Amazon S3 (Simple Storage Service):** Used for storing the raw data files and saving the generated query results.
- **Amazon Athena:** Used for creating databases and tables, and executing SQL queries directly on the data stored in S3 without the need to provision or manage servers.

3. Step-by-Step Execution Timeline

Phase 1: Data Storage Setup

1. **Bucket Creation:**
    - **Type:** General purpose.
    - **Namespace:** Account Regional namespace.
    - **Final Bucket Name:** healthcare-136876227305-us-east-1-an.
    - **Region:** us-east-1 (US East - N. Virginia).
2. **Data Ingestion:**
    - Uploaded a CSV data file named RealHealthcareData1.csv.
    - **File Size:** 1.1 KB.
    - **Storage Class:** Standard.

Phase 2: Data Cataloging (Database & Table Setup)

Using the Amazon Athena interface, the stored data was mapped to a queryable database through the following steps:

1. **Database:** Created a new database named healthcaredata.
2. **Table:** Created a table named patient_data.

3. **Dataset Location:** Pointed the table to read data from the S3 path: s3://healthcare-136876227305-us-east-1-an.
4. **Data Format:**
    - **Table type:** Apache Hive.
    - **File format:** CSV.
    - **SerDe library:** org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe.
    - **Field delimiter:** Comma ,.
5. **Schema Definition (Columns):** Defined 6 main columns:
    - Patient_ID (Data Type: int)
    - Name (Data Type: string)
    - Age (Data Type: int)
    - Diagnosis (Data Type: string)
    - Treatment (Data Type: string)
    - Files (Data Type: int)

Phase 3: Workgroup Configuration

Before executing queries, the primary workgroup settings were configured to ensure query results are managed and saved correctly:

- **Management of query results:** Selected "Athena managed" with default AWS-owned key encryption.
- **Location of query result:** Specified the exact same S3 bucket to store the output: s3://healthcare-136876227305-us-east-1-an.

Phase 4: Data Querying & Verification

To verify data integration and integrity, an SQL query was executed within the Athena Query Editor:

- **Executed Query:**

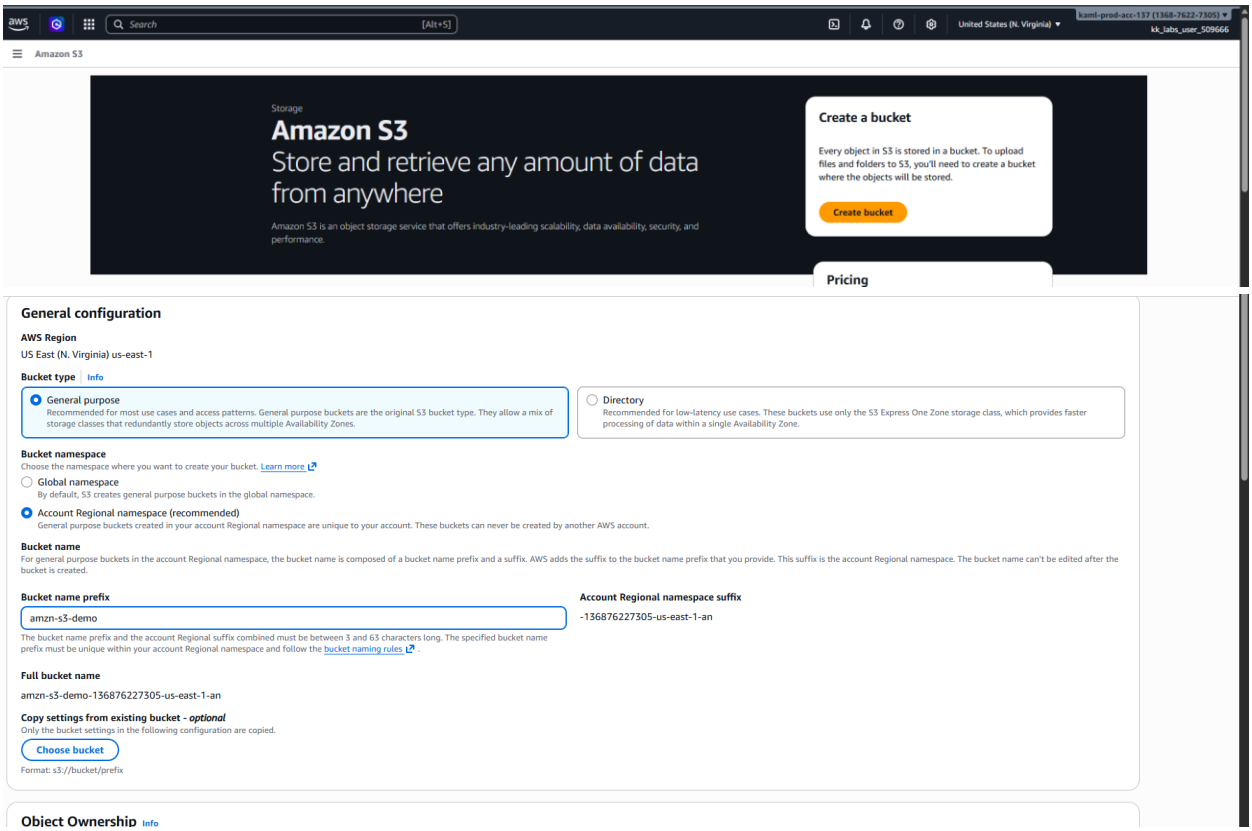
SQL

```
SELECT * FROM "healthcaredata"."patient_data" limit 10;
```

- **Execution Results:** The query completed successfully with a run time of 683 milliseconds, scanning 1.08 KB of data.
- **Output Analysis:** The results returned 10 rows of data.
    - *Technical Note:* The first row in the results displayed the original CSV text headers (e.g., "Diagnosis", "Treatment") as actual data values. This indicates that the current SerDe properties were not set to skip the first row (header row) of the source file. Subsequent records contained actual patient data (e.g., Thomas Clark, Emma Nelson) with medical diagnoses like Diabetes and Asthma.

4. Conclusion

Successfully built a streamlined, serverless data pipeline on AWS. The system is now fully capable of ingesting raw data files into S3 and enabling rapid, programmatic, or standard SQL-based analysis through Athena.



AWS Region
US East (N. Virginia) us-east-1

Bucket type Info

- General purpose
  Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.
- Directory
  Recommended for low-latency use cases. These buckets use only the S3 Express One Zone storage class, which provides faster processing of data within a single Availability Zone.

Bucket namespace
Choose the namespace where you want to create your bucket. Learn more

- Global namespace
  By default, S3 creates general purpose buckets in the global namespace.
- Account Regional namespace (recommended)
  General purpose buckets created in your account Regional namespace are unique to your account. These buckets can never be created by another AWS account.

Bucket name
For general purpose buckets in the account Regional namespace, the bucket name is composed of a bucket name prefix and a suffix. AWS adds the suffix to the bucket name prefix that you provide. This suffix is the account Regional namespace. The bucket name can't be edited after the bucket is created.

Bucket name prefix
healthcare

The bucket name prefix and the account Regional suffix combined must be between 3 and 63 characters long. The specified bucket name prefix must be unique within your account Regional namespace and follow the bucket naming rules.

Account Regional namespace suffix
-136876227305-us-east-1-an

Full bucket name
healthcare-136876227305-us-east-1-an

Copy settings from existing bucket - *optional*
Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Amazon S3 > Buckets > healthcare-136876227305-us-east-1-an > Upload

Upload Info

Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. Learn more

Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.

Files and folders (1 total, 1.1 KB)
All files and folders in this table will be uploaded.

Remove | Add files | Add folder

Find by name

Name	Folder	Type	Size
RealHealthcareData1.csv	-	text/csv	1.1 KB

Destination Info

Destination
s3://healthcare-136876227305-us-east-1-an

▸ **Destination details**
Bucket settings that impact new objects stored in the specified destination.

▸ **Permissions**
Grant public access and access to other AWS accounts.

▸ **Properties**
Specify storage class, encryption settings, tags, and more.

Cancel | Upload

healthcare-136876227305-us-east-1-an Info

Objects | Metadata | Properties | Permissions | Metrics | Management | File systems - *new* | Access Points

Objects (1)

Copy S3 URI | Copy URL | Download | Open | Delete | Actions | Create folder | Upload

Objects are the fundamental entities stored in Amazon S3. You can use Amazon S3 inventory to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. Learn more

Find objects by prefix

Name	Type	Last modified	Size	Storage class
RealHealthcareData1.csv	csv	April 23, 2026, 19:30:38 (UTC+02:00)	1.1 KB	Standard

aws | Search [Alt+S] | United States (N. Virginia) | kaml-prod-acc-137 (1368-7622-7305) kk_labs_user_509666

Analytics

Amazon Athena
Start querying data instantly.

Amazon Athena is an interactive query service that makes it easy to analyze data in Amazon S3 and other federated data sources using standard SQL.

Get started

- Query your data in Amazon SageMaker Unified Studio
  Run and schedule queries in a single data and AI development environment.
- Query your data in Athena console
  Use Query editor to analyze data on S3, on premises, or on other clouds.

Launch query editor

Pricing

Amazon Athena > Query editor

Typeahead suggestions are turned on by default. You can change this setting in query editor preferences.

Data

Data source

AwsDataCatalog

Database

Choose a database

Tables and views

Create

Filter tables and views

Tables (0)

Views (0)

Create a table from data source

S3 bucket data

AWS Glue Crawler

Create with SQL

CREATE TABLE

CREATE TABLE AS SELECT

CREATE TABLE AS SELECT(ICEBERG)

CREATE VIEW

Query 1

1

Explain

Cancel

Clear

Create

Query results

Query stats

Results

Table name

patient_data

Table name must be from 1-128 characters and must be unique. Valid characters are a-z, A-Z, 0-9, _(underscore). Table names tend to correspond to the directory where the data will be stored.

Description - *optional*

Type something

Table description must be from 1-1024 characters. 1024 characters remaining.

Database configuration Info

Choose an existing database or create a new database

Choose to access an existing database or to create a new database in order to create a new table. Athena stores the table schema in the AWS Glue Data Catalog.

- (●) Create a database
- ( ) Choose an existing database

Database name

Name your new database

healthcaredata

Database name must be from 1-128 characters and must be unique. Valid characters are a-z, A-Z, 0-9, _(underscore).

Dataset Info

Location of input data set

s3://healthcare-136876227305-us-east-1-an

View

Browse S3

Input the path to the data set you want to process on Amazon S3. For example if your data is stored at s3://input-data-set/logs/1.csv, please enter s3://input-data-set/logs/. If your data is already partitioned, e.g. s3://input-data-set/logs/year=2004/month=12/day=11/ just input the base path s3://input-data-set/logs/

Data format Info

Table type

Apache Hive

File format

CSV

SerDe library

org.apache.hadoop.hive.serde2.lazy.LazySimpleSerDe

SerDe properties - *optional*

Name	Value	
field.delim	,	Remove

Add SerDe property

Column details

Column name must be from 1-128 characters. Valid characters are a-z, A-Z, 0-9, _(underscore). Certain advanced column types (namely, structs) are not exposed in this interface.

Column name	Column type	Description - *optional*	
Patient_ID	int	Enter description	Remove
Name	string	Enter description	Remove
Age	int	Enter description	Remove
Diagnosis	string	Enter description	Remove
Treatment	string	Enter description	Remove
Files	int	Enter description	Remove

Add a column Bulk add columns

Amazon Athena > Workgroups

Simplify the administration of your query results using the managed query results
Have Athena store the query results for your workgroup securely and delete old query results for you after 24 hours. To use this setting, edit your workgroup and choose **Athena managed storage** for your query results location.

Workgroups (1) Info

Actions Create workgroup

Use workgroups to separate users, teams, applications, workloads, and to set limits on amount of data for each query or the entire workgroup process. You can also view query-related metrics in Amazon CloudWatch.

Filter workgroups Filter engine: Any engine

Name	State	Description	Analytics engine	Engine upd...	Authentication	Created on
primary	Turned on	-	Athena engine version 3	Automatic	IAM	2025-10-24T02:00:41.323+03:00

Amazon Athena > Workgroups > Edit primary

Authentication

- (selected) **AWS Identity and Access Management (IAM)**
  Authenticate using IAM to centrally manage permissions that control which AWS resources users can access.
- (disabled) **AWS IAM Identity Center**
  Authenticate with single sign-on using IAM Identity Center (recommended to centrally manage access permissions for multiple AWS accounts). You can add users once the workgroup is created.

⚠ You cannot change the authentication after the workgroup is created.

Query result configuration Info

Management of query results

- ( ) **Athena managed** - *new*
  Athena manages the storage for your queries and retains query results for 24 hours.
- (selected) **Customer managed**
  You manage the storage for your query results and keep the results as long as you require.

Location of query result - ***optional***
Enter an S3 prefix in the current region where the query result will be saved as an object.

s3://healthcare-136876227305-us-east-1-an [View] [Browse S3]

You can create and manage lifecycle rules for this bucket
Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. Learn more [Lifecycle configuration]

Expected bucket owner - ***optional***
Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

Enter AWS account ID

☐ Assign bucket owner full control over query results
Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.

☐ Encrypt query results

☐ Override client-side settings Info

Query result configuration Info

Management of query results

- (selected) **Athena managed** - *new*
  Athena manages the storage for your queries and retains query results for 24 hours.
- ( ) **Customer managed**
  You manage the storage for your query results and keep the results as long as you require.

Encrypt query results

- (selected) Encrypt using AWS owned key (Default)
  A key that AWS owns and manages for you.
- ( ) Encrypt using customer managed key
  Choose a key you have permission to use, or create a new one.

☐ Override client-side settings Info
If checked, workgroup settings apply to all queries in the workgroup.

Manage query result location and encryption

Location of query result - *optional*

Enter an S3 prefix in the current region where the query result will be saved as an object.

s3://healthcare-1368762273 | View | Browse S3

You can create and manage lifecycle rules for this bucket

Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time. Learn more

Lifecycle configuration

Expected bucket owner - *optional*

Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

Enter AWS account ID

☐ Assign bucket owner full control over query results

Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.

☐ Encrypt query results

Cancel | Save

Editor | Recent queries | Saved queries | Settings

Workgroup primary

Data

Data source: AwsDataCatalog

Database: healthcaredata

Tables and views | Create

Filter tables and views

Tables (1)

patient_data

Views (0)

Query 1 | Query 2 | Query 3

```
SELECT * FROM "healthcaredata"."patient_data" limit 10;
```

SQL Ln 1, Col 1

Running | Explain | Cancel | Clear | Create

Reuse query results up to 60 minutes ago

Query results | Query stats

Queued | Time in queue: 237 ms | Run time: - | Data scanned: -

Query results | Query stats

Completed — Time in queue: 111 ms — Run time: 683 ms — Data scanned: 1.08 KB

Results (10)

Copy | Download results

Search rows

1

#	patient_id	full_name	age	diagnosis	treatment	files
1		Name		Diagnosis	Treatment	
2	21	Thomas Clark	39	Diabetes	Diet control and metformin	2
3	22	Emma Nelson	57	Asthma	Bronchodilators	2
4	23	Oliver Scott	22	Anemia	Iron supplements	2
5	24	Charlotte Johnson	68	Arthritis	NSAIDs	2
6	25	Liam Wright	33	Depression	Cognitive behavioral therapy	2
7	26	Isabella Martin	49	Skin Cancer	Surgical removal	2
8	27	Ethan King	28	Acid Reflux	Proton pump inhibitors	2
9	28	Ava Lee	41	Obesity	Bariatric surgery	2